BESS Optimal Arbitrage Report – Germany (DE)

1 Basic Inference

The BESS charges during periods of lowest prices (e.g., 0.0097 €/kWh at 13:00 on 2025-06-04) and discharges during peak prices (e.g., 0.1929 €/kWh at 20:00 on 2025-06-04). For c = 0.5 (5 kW/10 kWh), the profit for 2025-06-04 was **1.47 €**, with a charge cost of **0.14€** and discharge revenue of **1.61€**.

2 Seasonality Effect

Price peaks occurred during evenings and lower prices during midday, probably due to high availability of renewable energies, which the BESS exploits for arbitrage. It is to be assumed, that high availability of renewable energies lowers the prices due to overproduction that cannot be controlled. During periods with a lower availability of renewable energies, prices will spike less, which will lead to a lower profit margin.

3 Single FCE Constraint

If the BESS is limited to one full cycle per day, the optimal strategy is to charge fully during the lowest price hours and discharge during the highest price hours.

3.2 Profit Change with Various C-Rates

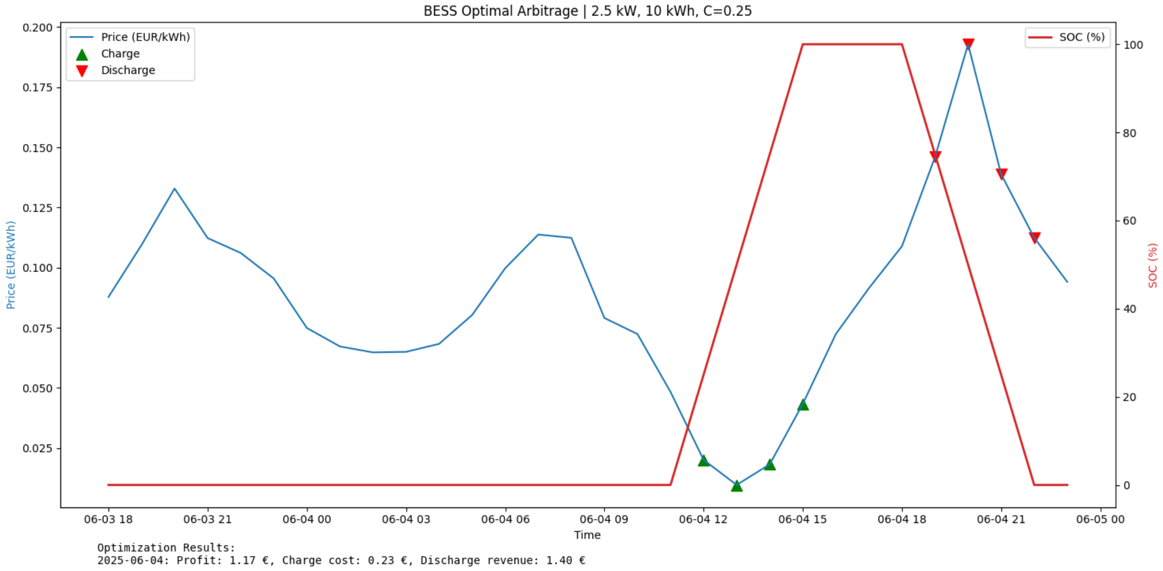
C-Rate (kW/10 kWh)	Max Power (kW)	Profit (€/day, 2025-06-04)
0.25	2.5	1.17
0.50	5.0	1.47
0.75	7.5	1.47
1.00	10.0	1.74

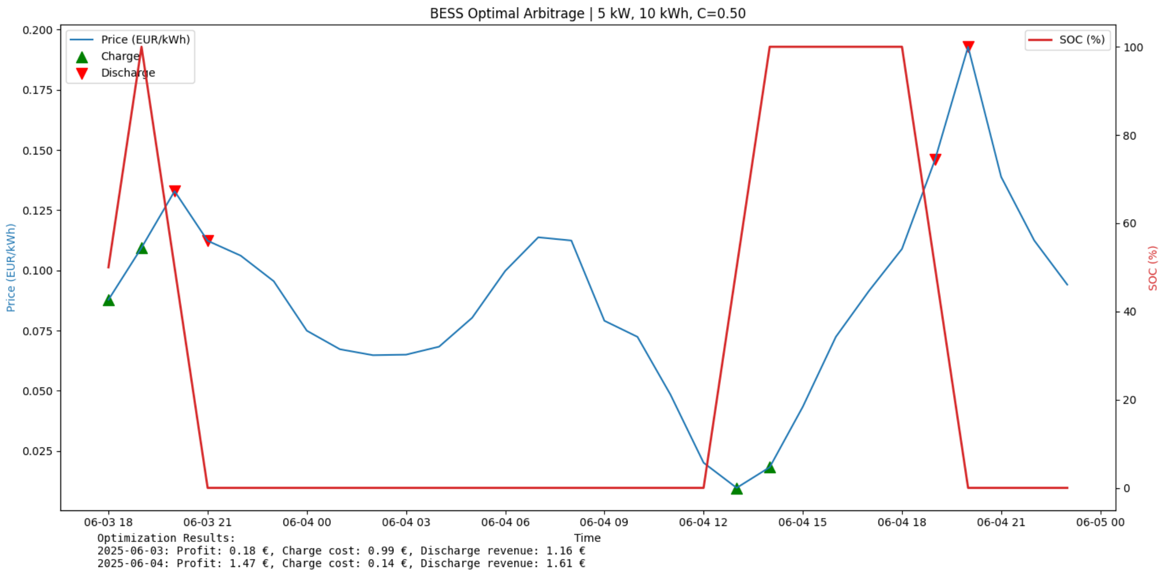
Observation:
Higher C-rates allow faster response to price spikes, increasing potential profit, but the gain saturates as the BESS can already exploit the main price differential at c = 0.5–0.75.

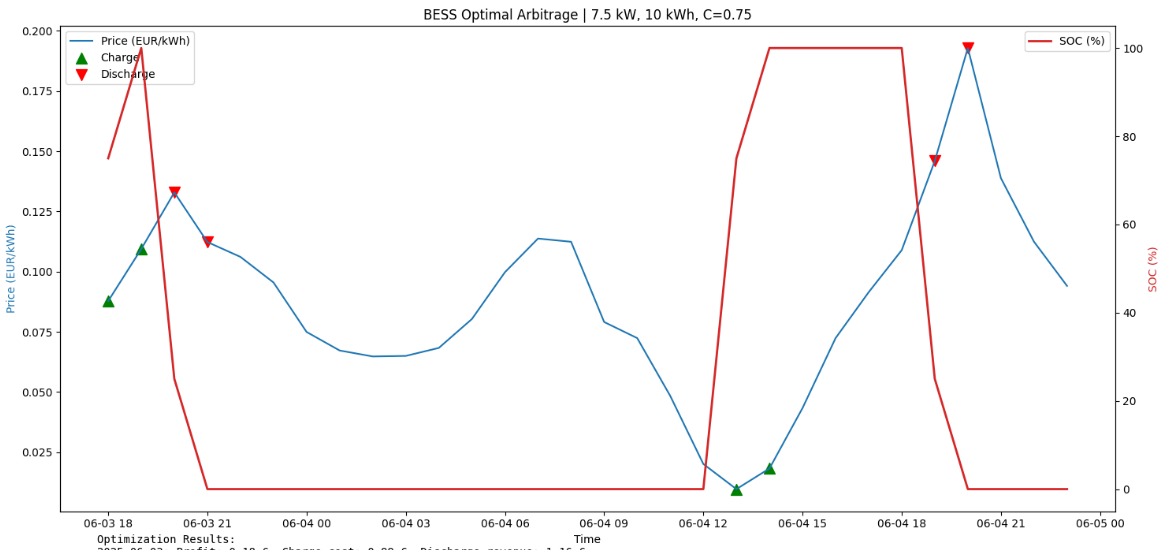
4 Conclusion

- BESS arbitrage on the German Day-Ahead Market is profitable under current price conditions, with profit increasing with higher C-rates and optimal timing.
- The most significant gains come from exploiting the largest price differences, typically between midday lows and evening highs.

Data Visualization





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Optimization Results:
2025-06-03: Profit: 0.18 €, Charge cost: 0.99 €, Discharge revenue: 1.16 €
2025-06-04: Profit: 1.47 €, Charge cost: 0.14 €, Discharge revenue: 1.61 €
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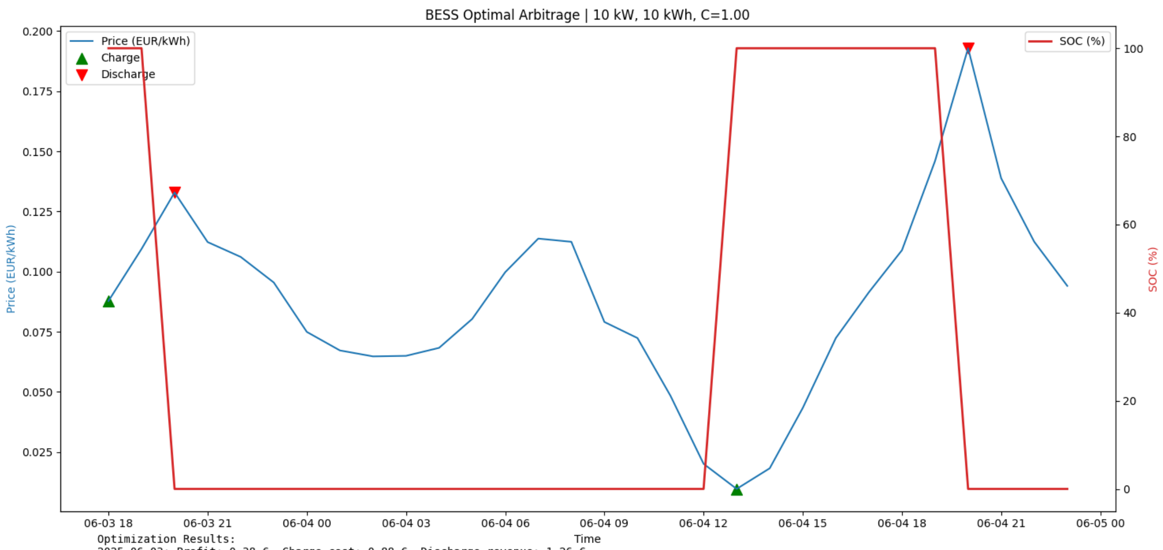


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Optimization Results:
2025-06-03: Profit: 0.38 €, Charge cost: 0.88 €, Discharge revenue: 1.26 €
2025-06-04: Profit: 1.74 €, Charge cost: 0.10 €, Discharge revenue: 1.83 €
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